

400 V – 24 V / 10.5 A DC-DC resonant LLC converter for industrial applications, using MasterGaN1L, L6599A, and SRK2001A



Introduction

The **EVL250WMG1L** evaluation board is a DC-DC resonant LLC converter, dedicated to any kind of industrial application where minimum size and high efficiency are required (industrial DC-DC applications, adapters, consumer SMPS). The half-bridge of the converter is based on the MasterGaN1L, embedding a couple of GaN power transistors and a driver in the same package, allowing to directly interface any kind of SMPS controller. Thanks to the GaN technology and to the embedded driver, the converter can be designed with an operating frequency higher than using conventional MOSFETs, allowing the design of a smaller transformer. Actually, the board has no heat sink on the primary side and has very reduced dimensions. The high efficiency and small size make the board very suitable when available space is limited; the power density is 34 W/inch³. Output power can be up to 250 W at 24 Vdc and the nominal input voltage is 400 V, as delivered by conventional front-end PFCs. The board comes with overcurrent, shot-circuit, and voltage loop fail protection. The integrated input voltage monitoring allows the LLC converter to start up with correct sequencing, preventing operation with too low input voltage. The primary side controller is L6599A, while the secondary side synchronous rectification is based on SRK2001A. An image of the evaluation board is shown in [Figure 1](#), while its block diagram is shown in [Figure 2](#).

Figure 1. EVL250WMG1L evaluation board

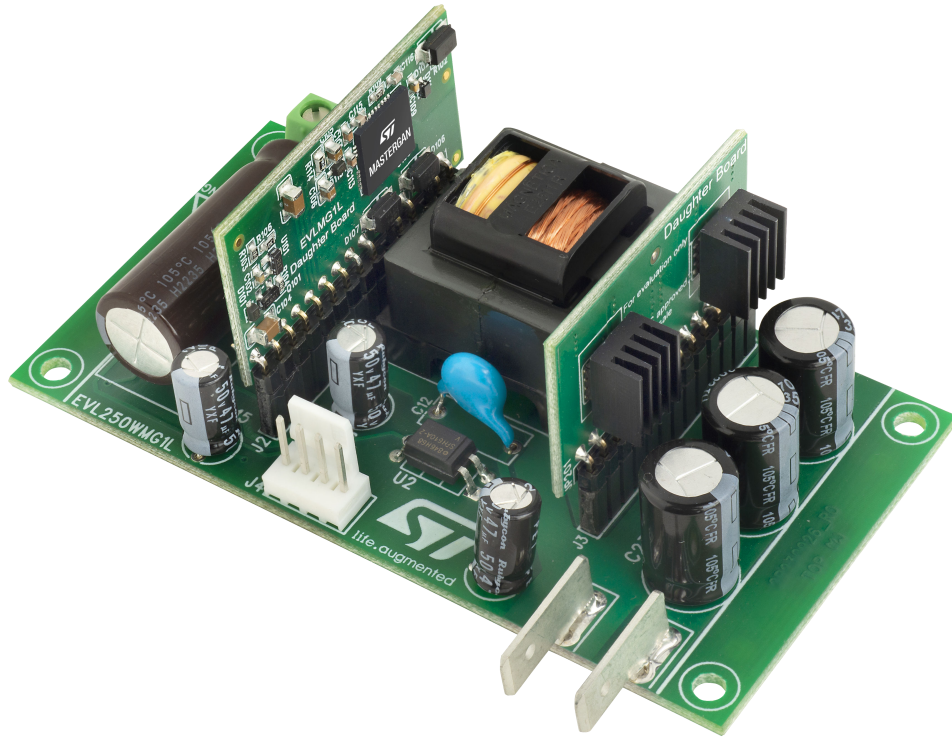
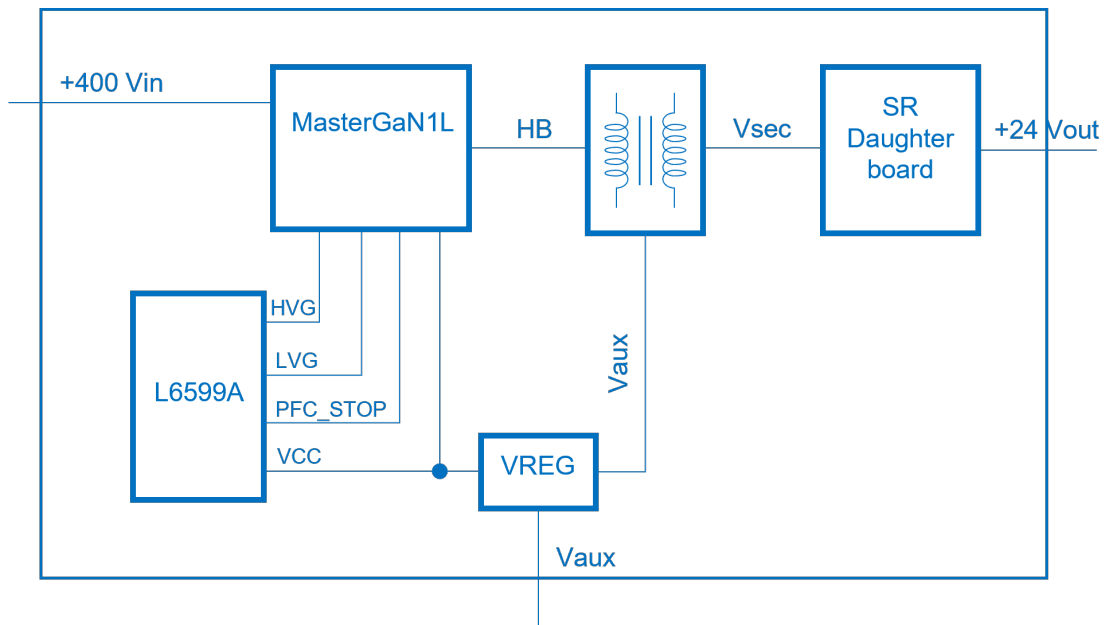


Figure 2. EVL250WMG1L: functional block diagram



1 EVL250WMG1L schematic

Figure 3. EVL250WMG1L motherboard schematic

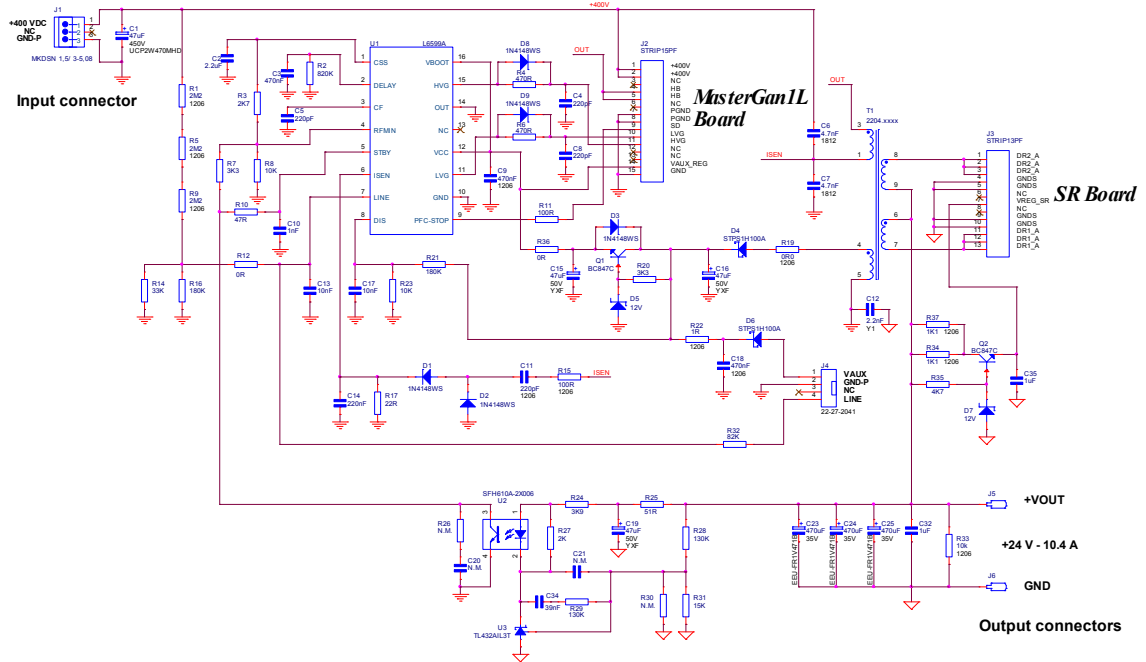


Figure 4. EVL250WMG1L MasterGaN1L module schematic

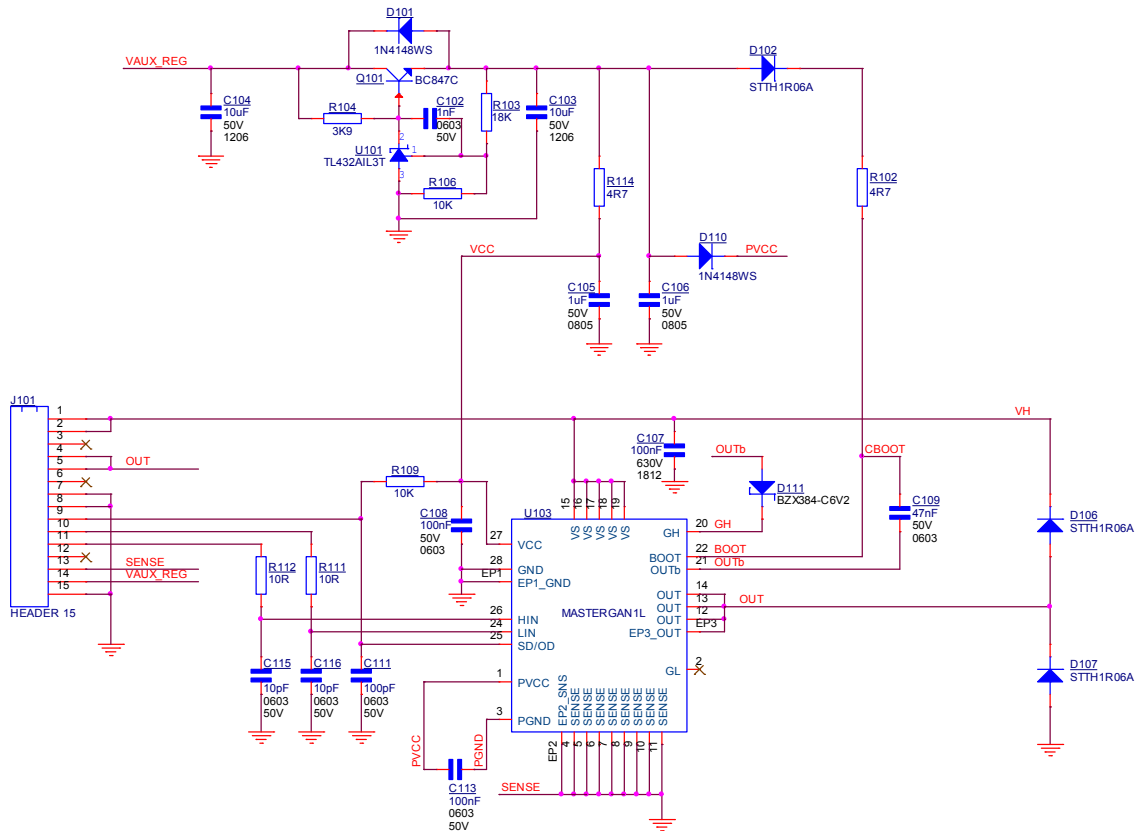
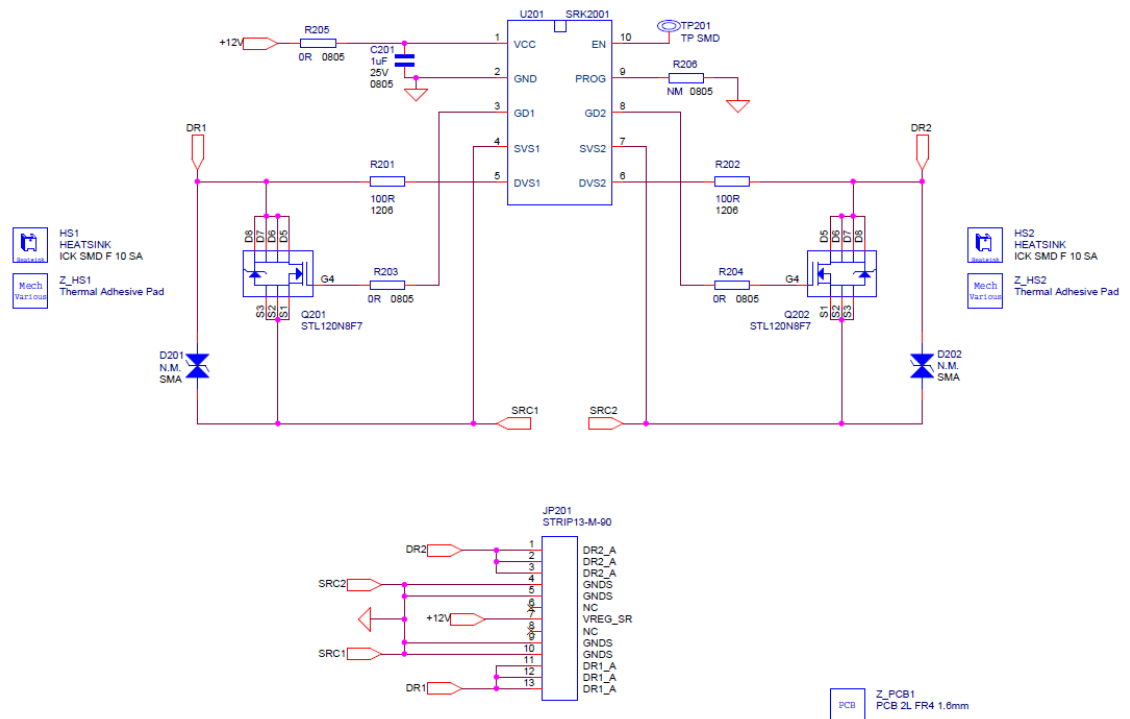


Figure 5. EVL250WMG1L SRK module schematic



2 EVL250WMG1L PCB layout

Figure 6. EVL250WMG1L motherboard PCB tracks - top side

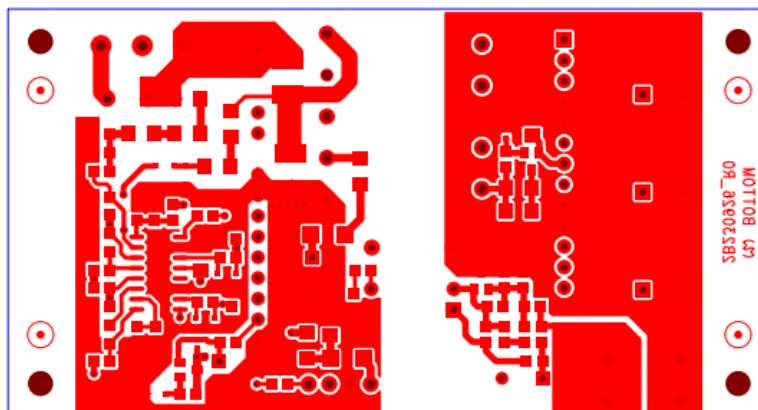


Figure 7. EVL250WMG1L motherboard PCB tracks - bottom side

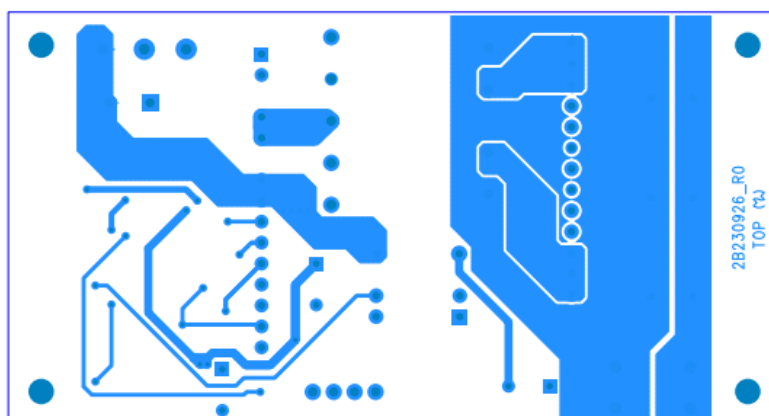


Figure 8. EVL250WMG1L motherboard PCB layout - top side

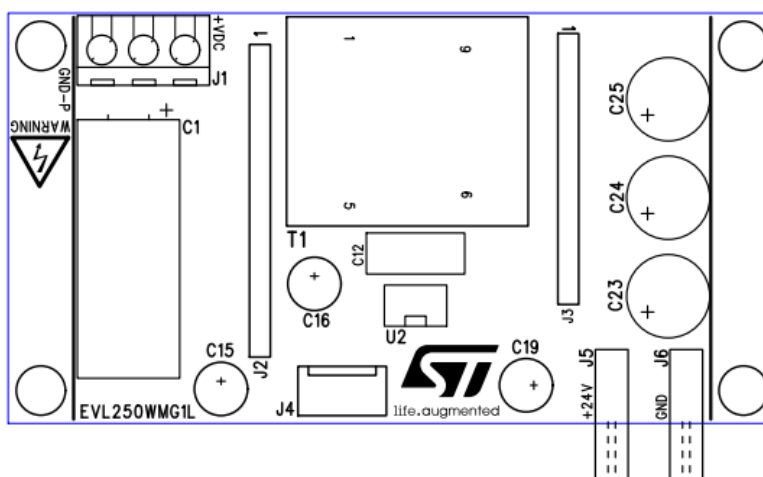


Figure 9. EVL250WMG1L motherboard PCB layout - bottom side

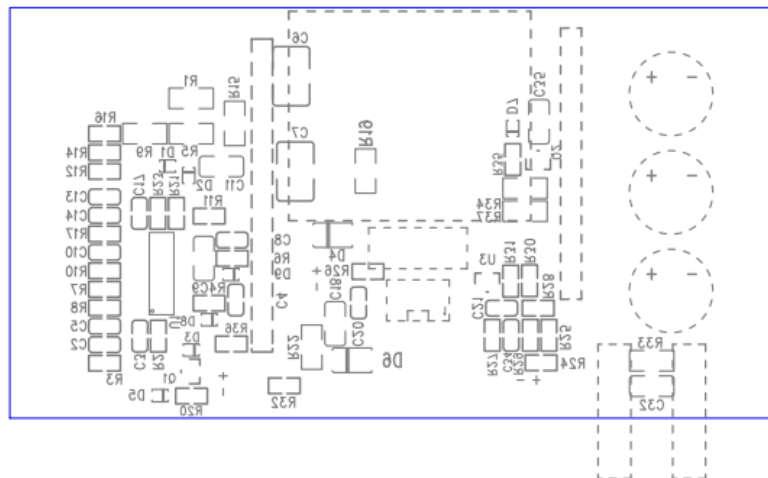


Figure 10. MasterGaN1L daughterboard PCB tracks - top side

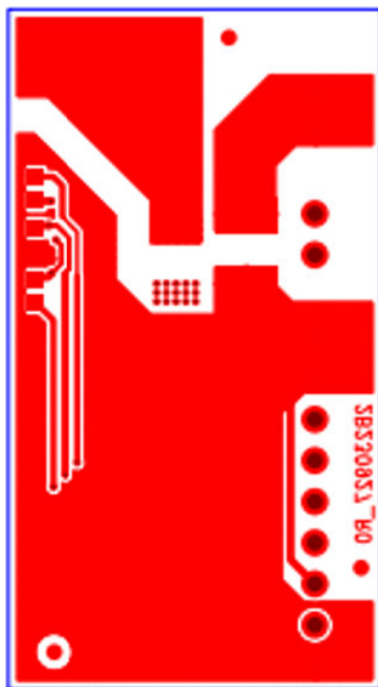


Figure 11. MasterGaN1L daughterboard PCB tracks - bottom side

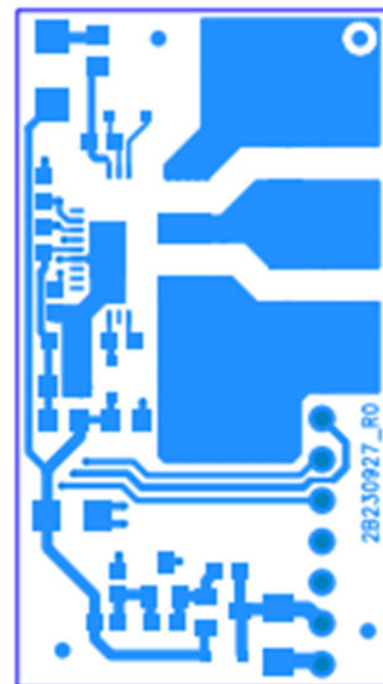


Figure 12. MasterGaN1L daughterboard PCB layout - top side

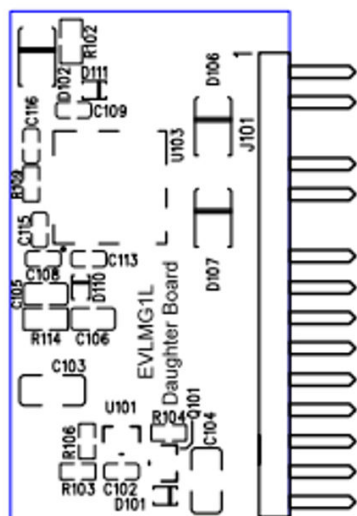


Figure 13. MasterGaN1L daughterboard PCB layout - bottom side

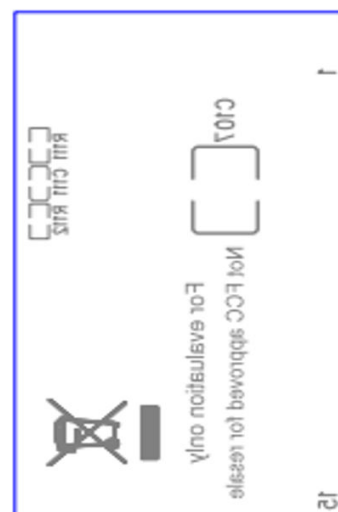


Figure 14. Synchronous rectifier daughterboard PCB tracks - top side

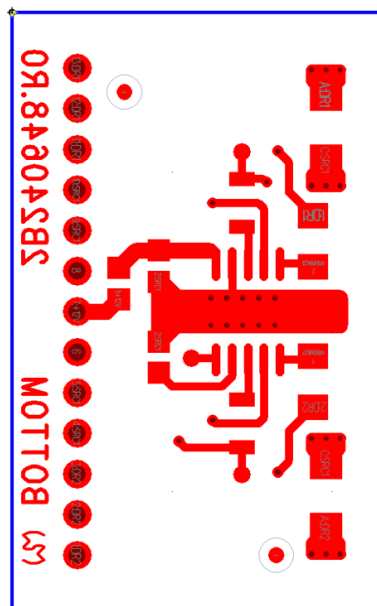
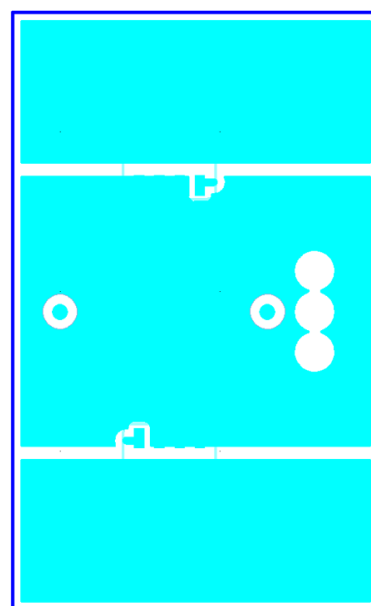
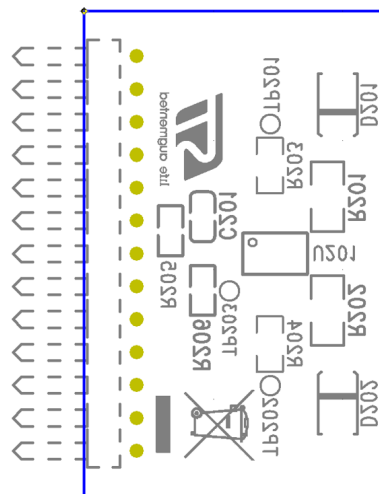


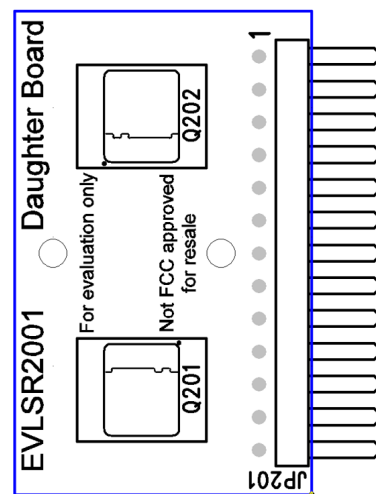
Figure 15. Synchronous rectifier daughterboard PCB tracks - bottom side



**Figure 16. Synchronous rectifier daughterboard
PCB layout - top side**



**Figure 17. Synchronous rectifier daughterboard
PCB layout - bottom side**



3 EVL250WMG1L description

The EVL250WMG1L evaluation board is composed of one motherboard (with L6599A, the LLC tank, and the secondary side circuitry apart from the synchronous rectification stage) and two daughterboards (one for MasterGaN1L and one for the secondary side synchronous rectifier) plugged onto it. The EVL250WMG1L is supplied by 400 Vdc (J6) as power rail; an additional 15 V referred to primary GND has to be provided to supply the control part on connector J3. This auxiliary voltage is needed just for startup; after startup the external auxiliary can be removed since the Vcc is self supplied via the primary transformer. The board is based on a resonant LLC topology, the working frequency at full load is around 260 kHz, and output power is 250 W at 24 Vdc.

The power transformer is driven by the MasterGaN1L, an advanced power system-in-package integrating a gate driver and two enhancement mode GaN FET connected in half-bridge configurations needed by the selected topology. The MasterGaN1L is supplied by a linear regulator delivering 6.9 V, composed of Q101, U101, R103, and R106. This regulator voltage is used to supply the MasterGaN1L VCC pin and, via the diodes D102 and D110, the drivers of the low-side and the high-side FET. The control IC is the L6599A, a double-ended controller dedicated to resonant half-bridge topology. It provides two output signals with 50% complementary duty cycle, 180° out-of-phase with a fixed dead time inserted between the turn-off of one switch and the turn-on of the other one allowing the ZVS operation.

The L6599A allows to set the minimum T and maximum operating frequency by R7, R8, and C2; the STBY pin allows to set the burst mode threshold. In order to have the LLC stage activation with proper input voltage, the L6599A, by means of a driver R1, R5, R9, R14, R16 connected to the LINE pin, monitors the converter input voltage preventing operation with too low input voltage. Additional protections are OCP and output voltage loop fail, managed by pins ISEN and DIS.

To achieve high efficiency the board uses synchronous rectification on the secondary side, managed by SRK2001A driving the MOSFETs Q201 and Q202. This portion is located on the daughterboard located on the secondary side. A linear voltage regulator on the motherboard bottom side scales down the output voltage to 12 V, to supply the SRK2001A.

4 EVL250WMG1L board overview

Table 1 and Table 2 below list the EVL250WMG1L characteristics and connectors.

Table 1. EVL250WMG1L characteristics

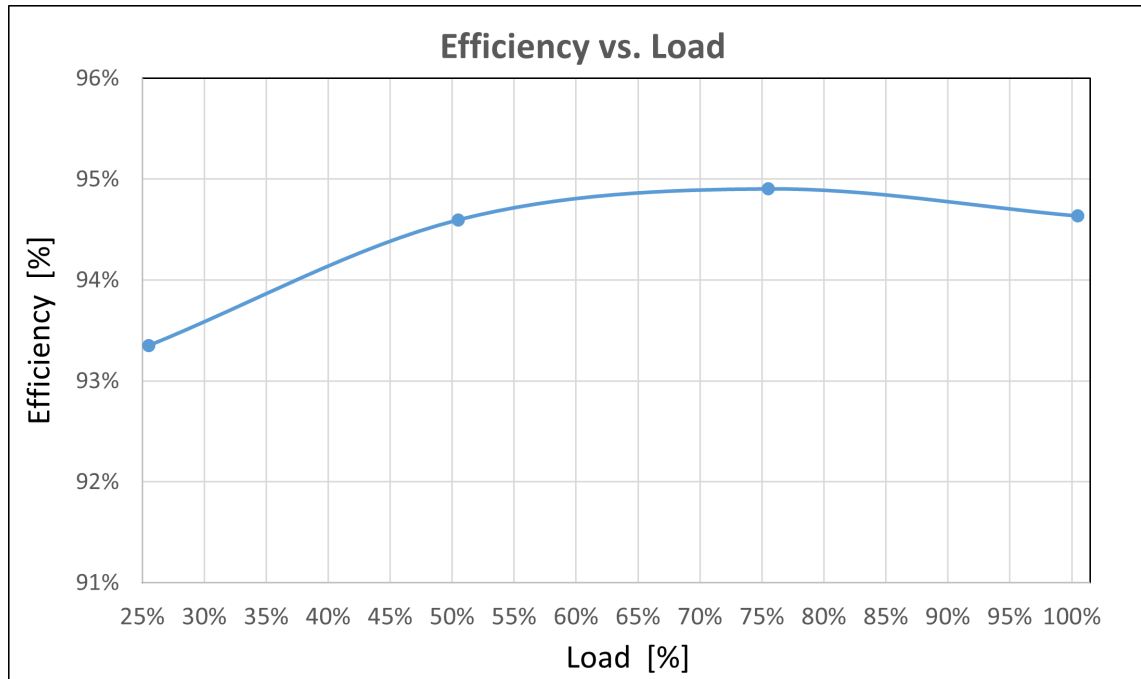
Parameter	Value
DC input voltage	400 V $\pm$ 10%
DC output voltage	24 V
Output current	Up to 10.4 A
Output power	P _{out} = 250 W
Efficiency	Eff. > 94% at full load
No load consumption	$\leq$ 375 mW
Operating frequency @ full load 400 Vin	260 $\pm$ 10 kHz
Board size	80 x 50 (WxH) mm Maximum component height: 30 mm

Table 2. EVL250WMG1L connectors

Reference	Type	Description
J1	PCB terminal	Power input
J4	PCB headers	Auxiliary input voltage
J5	Faston terminal (+)	Power output
J6	Faston terminal (-)	Power output

5 EVL250WMG1L performance and efficiency measurement

Figure 18. EVL250WMG1L efficiency



The above diagram shows the efficiency measured from a 25% to 100% load. At maximum output power the measured efficiency is over 94%.

6 EVL250WMG1L waveforms

Figure 19 and Figure 20 show the startup operation in both full load and no load conditions.

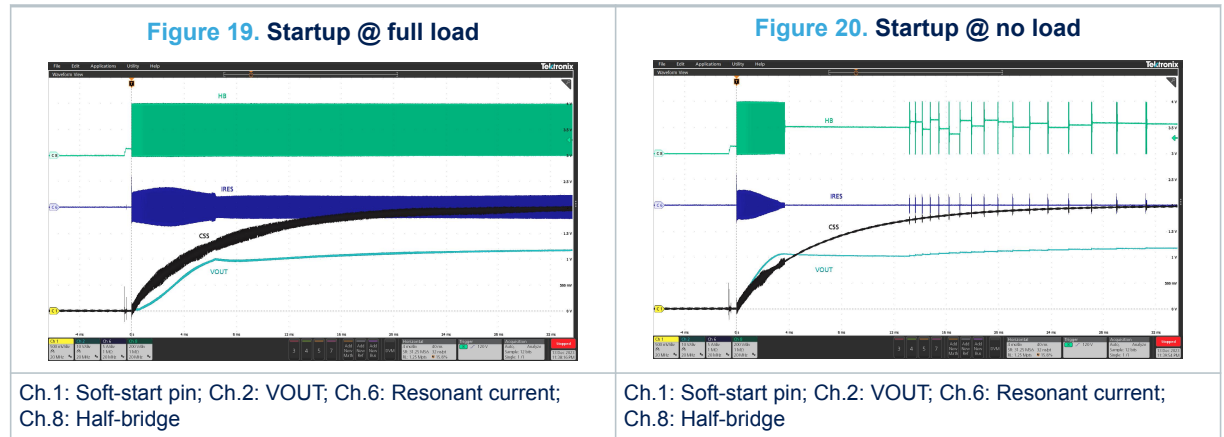


Figure 21 and Figure 22 show the steady-state operations at 8 A and no load conditions (burst mode).

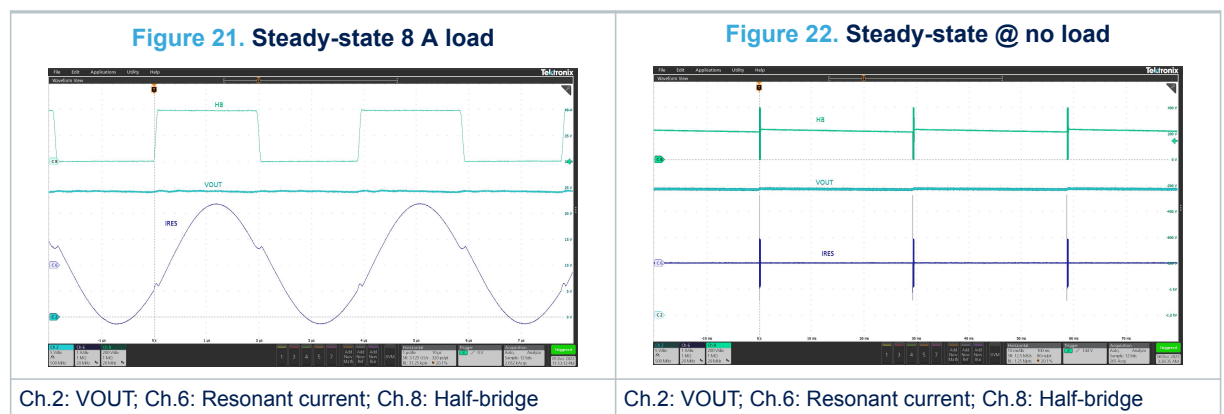


Figure 23 and Figure 24 show the steady-state conditions at full and half load: HVG and LVG signals from the L6599A are shown as reference.

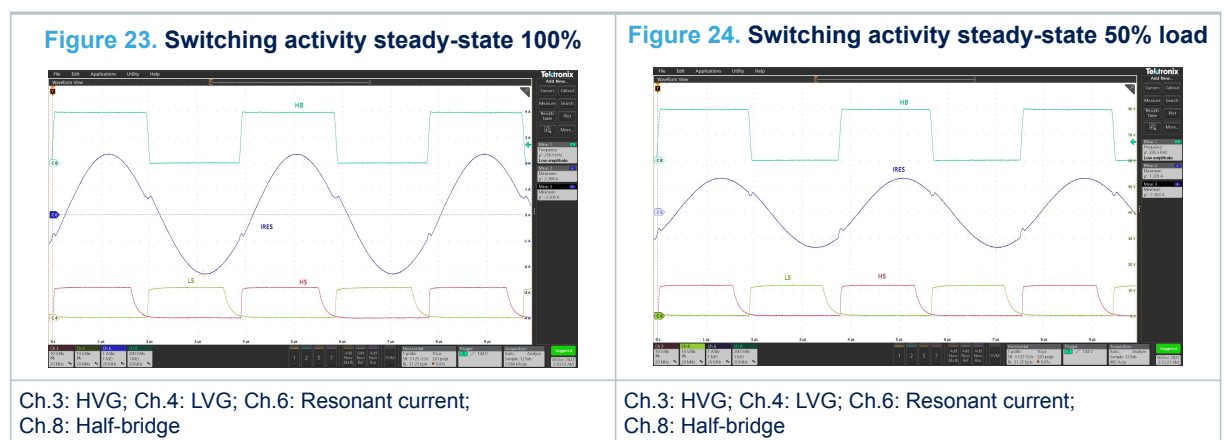
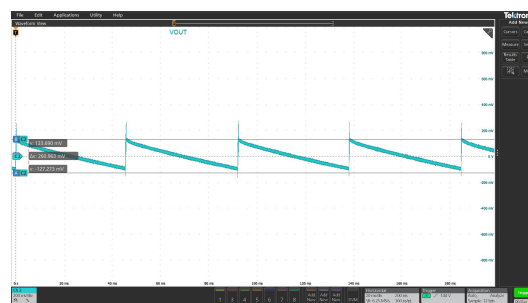


Figure 25 and Figure 26 show the output voltage ripple. The Vpeak to peak ripple are 120 mV at full load and 500 mV at no load during burst mode operation.

Figure 25. Vout ripple full load

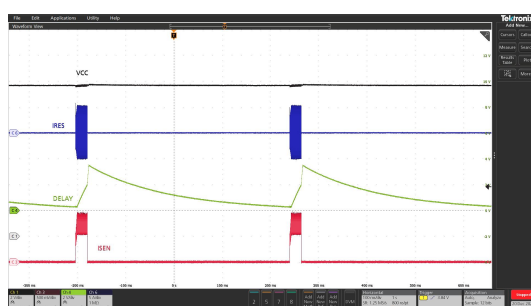

Ch.2: VOUT ripple

Figure 26. Vout ripple no load


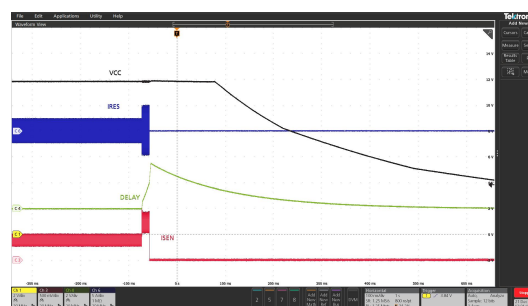
Ch.2: VOUT ripple

The EVL250WMG1L evaluation board is protected by short-circuit or overcurrent that might occur, preventing components overheating or catastrophic failures of the board. **Figure 27** and **Figure 28** show the safe interruption of switching activity. In **Figure 27**, since the board is supplied by an external Vcc, it can be observed that the board starts working in hiccup mode until the short is removed. Then it restarts via a soft-start cycle.

In **Figure 28**, since the external Vcc is removed and the board is self-supplied, we can see that after the short is applied the converter works for a short time and then stops the operation because the Vcc is lost and the converter cannot restart. To resume operation, it is necessary to reconnect the external Vcc allowing the board startup.

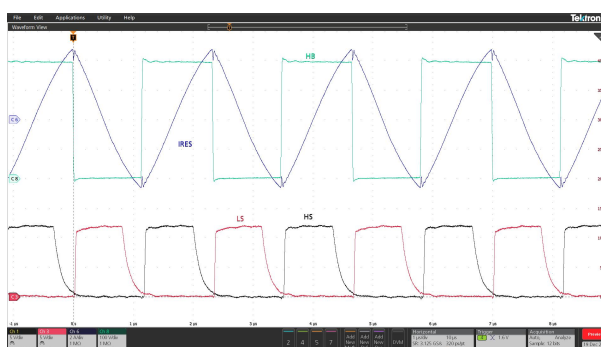
Figure 27. Overcurrent protection with ext. VCC


Ch.1: External VCC; Ch.3: ISEN pin; Ch.4: Delay pin; Ch.6: Resonant current

Figure 28. Overcurrent protection without ext. VCC


Ch.1: External VCC; Ch.3: ISEN pin; Ch.4: Delay pin; Ch.6: Resonant current

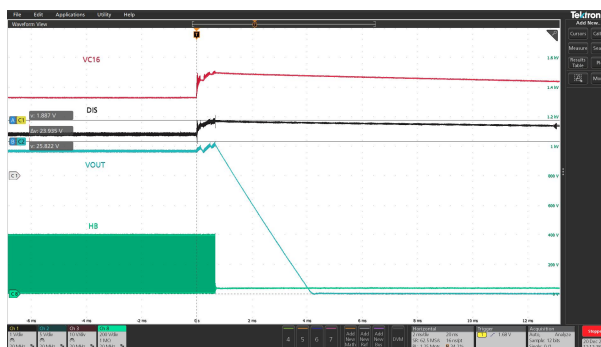
Figure 29 is a zoom in during the switching phase of the hiccup operation, focused on the driving signals of the half bridge, the half bridge voltage and the resonant current.

Figure 29. OCP switching activity detail


Ch.1: HVG; Ch.3: LVG; Ch.6: Resonant current; Ch.8: Half-bridge

An additional protection of the EVL250WMG1L evaluation board is the open loop failure disconnection, sensing the auxiliary voltage. As shown in [Figure 30](#), in case of an output voltage loop fail, the auxiliary voltage V_{aux} rises and once the voltage on the DIS pin has reached the 1.88 V threshold, the L6599A stops operation and latches the switching activity. The converter operation is resumed after the V_{cc} has dropped below the UVLO threshold. The measured V_{out} is 25.8 V, below the maximum voltage rating of the output capacitors.

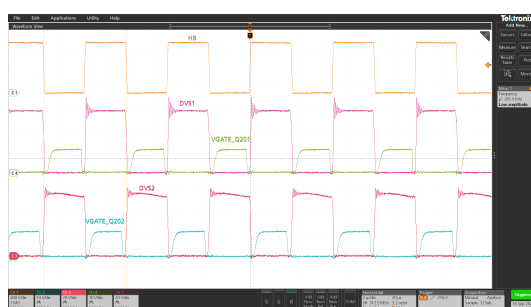
Figure 30. Overvoltage protection in case of feedback failure disconnection



Ch.1: L6599A DIS pin; Ch.2: VOUT; Ch.3: VC16; Ch.8: Half-bridge

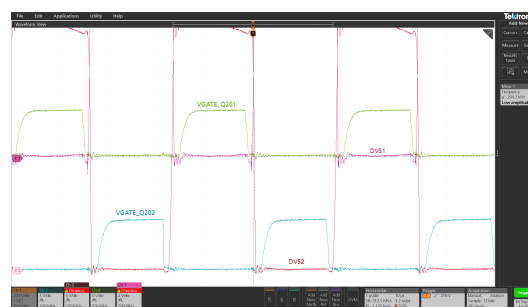
[Figure 31](#) shows the drain and gate voltages of secondary MOSFETs driven by the SRK2001A, synchronous rectifier controller. [Figure 32](#) shows the detail of the secondary MOSFET's drain voltage. The voltage drop variation before and after each MOSFET turn-on and turn-off.

Figure 31. SR waveforms



Ch.1: Half bridge; Ch.2: V_{GATE_Q202} ; Ch.3: DVS2; Ch.4: V_{GATE_Q201} ; Ch.7: DVS1

Figure 32. SR drain voltage detail



Ch.2: V_{GATE_Q202} ; Ch.3: DVS2; Ch.4: V_{GATE_Q201} ; Ch.7: DVS1

7 EVL250WMG1L thermal map

In Figure 33 the EVL250WMG1L evaluation board thermal map at full load is shown.

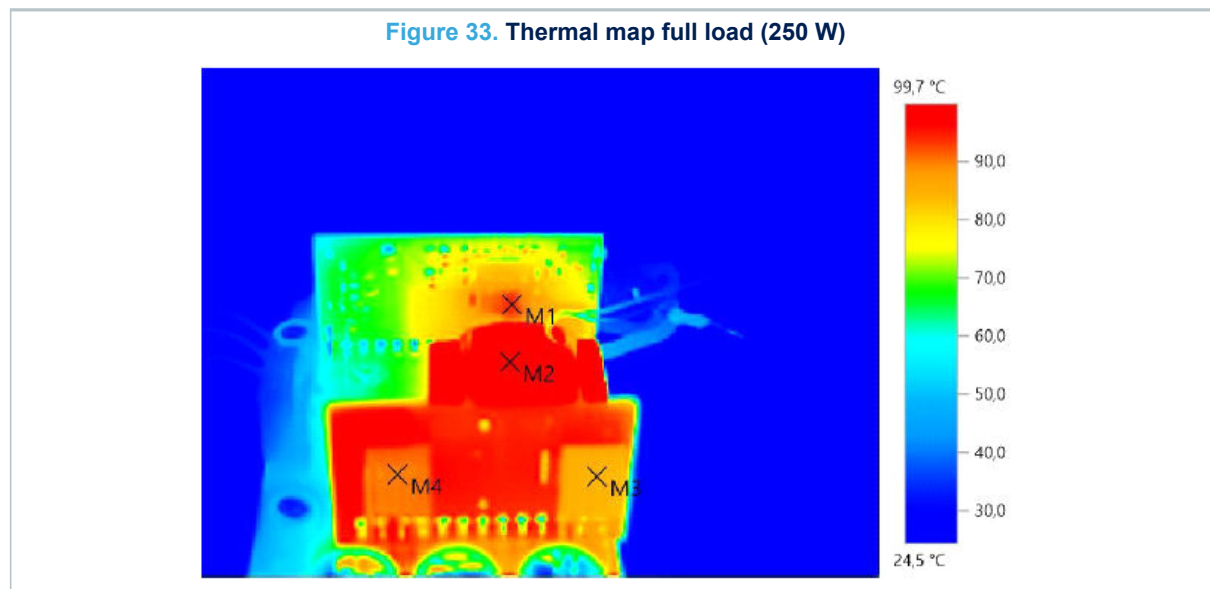


Table 3. Thermal map points description

M1	MasterGaN1L	94.0 °C
M2	Primary transformer	100.0 °C
M3	Heat sink secondary MOSFET	83.7 °C
M4	Heat sink secondary MOSFET	89.5 °C

8 References

MasterGaN1L datasheet: DS14355

L6599A Datasheet: DS6027

SRK2001A Datasheet: DS11726

Revision history

Table 4. Document revision history

Date	Version	Changes
18-Sep-2024	1	Initial release.

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